

Demonstration of Proposed Features

Date	4 july 2026
Team ID	XXXXXX
Project Name	OPTICROP- Smart Agricultural Production Optimization Engine"
Maximum Marks	1 Mark

Demonstration of Proposed Features:

This document captures all the features that were proposed during the project planning phase and tracks whether each feature was successfully implemented and demonstrated. It serves as evidence of the team's ability to deliver on the proposed solution.

S.No	Feature Name	Description	Status (Implemented / Partial / Pending)	Demonstrated (Yes / No)	Remarks
1	Smart Crop Recommendation Engine	Accepts inputs (N,P,K , temperature, humidity, pH, and rainfall) from the user and outputs the ideal crop recommendation using the trained ML model.	implemented	Yes	Core algorithm achieved high validation accuracy during testing.
2	Interactive User Interface (UI)	A clean, responsive web form built with HTML/CSS and served via Flask to allow farmers to easily input data and view results.	Implemented	Yes	Seamless navigation across different user application scenarios.
3	Crop Suitability Assessment	Evaluates if current conditions match a specific, pre-selected crop and provides feedback on productivity compatibility.	Implemented	Yes	Allows specific testing for targeted agricultural planning.
4	Explanatory Data Visualizations	Interactive dashboards using Matplotlib/Seaborn to display crop-environment trends and historical analysis for researchers.	partial	Yes	Basic trends and feature correlations are visible; advanced filtering is in progress.
5	Robust Data Pre-processing Pipeline	automated scaling, normalization, and handling of outliers to ensure clean data reaches the machine learning model.	implemented	yes	Handles empty or anomalous user inputs gracefully to prevent app crashes.
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Feature Implementation Summary:

Total Features Proposed	✓ Total Features Proposed: 5
Total Features Implemented	Total Features Implemented: 4 (Full) + 1 (Partial)
Total Features Demonstrated	Total Features Demonstrated: 5
Overall Implementation Rate (%)	Overall Implementation Rate (%): 90%